Purpose of the project

This project aims to investigate whether this banknote authentication dataset is suitable to serve as the basis to distinguish between genuine and forged banknotes.

Data resource

A special tool, Wavelet Transform, was used to extract features from the images taken from both genuine and forged banknotes. Then a number of features were recorded, including variance, skewness, kurtosis, and entropy of the images. In the supplied dataset (in CSV format): there are 1372 instances of 2-dimensional observations (V1: variance and V2: skewness).

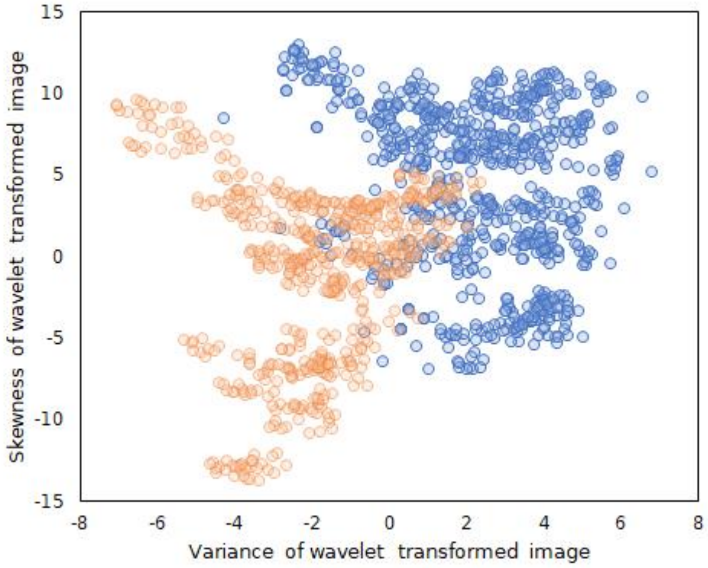
V1	V2
3.6216	8.6661
4.5459	8.1674
3.866	-2.6383
3.4566	9.5228
0.32924	-4.4552
4.3684	9.6718
3.5912	3.0129
2.0922	-6.81
3.2032	5.7588
1.5356	9.1772
1.2247	8.7779

V1 (Variance): • Range: 0.43 to 6.82 • Mean: 0.43373 • Standard Deviation: 2.84172

V2 (skewness): • Range: 1.92 to 12.95 • Mean: 1.92235 • Standard Deviation: 5.86691

Terminology

We would be applying K-means clustering with the number of clusters as two (i.e.: separating banknotes in two groups), and compare to the identified classes in the full dataset to check if the model is accurate enough to be used for forged and genuine notes according to these features.



Result

There are two clusters that represent two notes: one authentical note and the other one needs us to identify. If the second one is also authentical, it must meet the below requirements:

The cluster must completely identical to the authentical note, which including the number of points, the location of points. From the graph we can tell that: they share a very similar shape, but not exactly. We can see one cluster slides away. Second, the corresponding locations are different.

Conclusion

Based on the above information, we consider the note is forged. Even though the appearance is very similar, however, because standard deviation enlarged the difference of the location.

Recommendations for client

Data science is a very efficient way to identify the forged note. We can introduce more variables into the identification technology in order to have a more noticeable result.

There could be several different versions of banknotes, which have more characters, not only the values and SD, such as serial numbers, printing technologies, etc. That would make forging a note even more difficult.